22.2 Applications of Microbes

Microorganisms contribute to human welfare in many ways. In this section we will study about the diversified usefulness of microbes.

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22.2.1 Microbes in Agriculture

Microbes play an important role in agriculture as biocontrol agents and biofertilizers. Microbes play a vital role in the cycling of elements like carbon, nitrogen, oxygen, sulphur and phosphorus. These are called biological scavangers.

Microbes as biofertilizers: Microorganisms which enrich the soil with nutrients are called as biofertilizers. Bacteria, cyanobacteria and fungi are the main sources of biofertilizers. Nitrogen is one of the main source of plant nutrients. Atmospheric nitrogen has to be converted to available form of nitrogen. This is done by microbes either in free living conditions or by having symbiotic relationship with the plants. e.g. *Azotobacter*, Nitrosomonas *Nostoc* (free living), symbiotic microbes like *Rhizobium*, *Frankia*.

Activity 1

Take the root nodules of any pulse or leguminous plant available in your locality. Wash it throughly with water. Crush the nodules on a clean glass slide. Add a drop of distilled water to the crushed material on the glass slide. Observe the preparation under compound microscope.

Microbes as biocontrol agents: Microorganisms used for controlling harmful or pathogenic organisms and pests of plants are called as biocontrol agents (Biopesticides). *Bacillus thuringiensis* (Bt) is a species of bacteria that produces a protein called as 'cry' protein. This protein is toxic to the insect larva and kills them. Spores of *B.thuringiensis* are available in sachets, which are dissolved in water and sprayed on plants infected with insect larva.

22.2.2 Microbes in Industries

Microorganisms play an important role in the production of wide variety of valuable products for the welfare of human beings.

Production of fermented beverages: Beverages like wine are produced by fermentation of grape fruits by *Saccharomyces cerevisiae*.

Curing of coffee beans, tea leaves and tobacco leaves: Beans of coffee and cocoa, leaves of tea and tobacco are fermented by the bacteria *Bacillus megaterium*. This gives the special aroma.

Production of curd: *Lactobacillus sp.* converts milk to curd.

Production of organic acids, enzymes and vitamins: Oxalic acid, acetic acid and citric acid are produced by fungus *Aspergillus niger*. Enzymes like lipases, invertase, proteases, and glucose oxidase are derived from microbes. Yeasts are rich source of vitamin-B complex.

22.2.3 Microbes in Medicine

Antibiotics: These are metabolic products of microorganisms, which in very low concentration are inhibitory or detrimental to other microbes. In 1929, Alexander Fleming produced the first antibiotic pencillin. In human beings antibiotics are used to control infectious diseases like cholera, diptheria, pneumonia, typhoid, etc.

Table 22.1 Antibiotics produced by micro organisms

Class of Microorganisms	Type of Microorganism	Antibiotic produced
Bacteria	*Streptomyces griseus*	Streptomycin
	Streptomyces erythreus	Erythromycin
	Bacillus subtilis	Bacitracin
Fungi	*Penicillium notatum*	Penicillin
	Cephalosporium acremonium	Cephalosporin

Vaccines: These are prepared by killing or making the microbes inactive (attenuated). These inactive microbes are unable to cause

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